

NASA Mission-Map Animation vs. Our Telemetry — Entry Point → Splashdown

A truth-mode, side-by-side forensic comparison

Scope of this document. It compares exactly two things, and nothing else:

- 1. **NASA’s own animation** — the file `jsc2025m000169_Artemis_II_Mission_Map_4K.mp4` , specifically the last-day re-entry segment (Entry Point → Splashdown), as it visually appears on screen.
- 2. **Our telemetry** — `traj_telemetry.json` , 265 frames read directly off the on-screen dials of the broadcast we captured.

Everything below is separated into **PROVABLE** (directly visible in one of those two sources) and **AS-SUMED / UNKNOWN** (not present in either, or inferred). No outside data was consulted. No number was invented. Where the two sources disagree or cannot check each other, that is stated plainly.

1. The two sources are not the same kind of thing

	NASA mission-map anima-tion	Our captured telemetry
What it is	A stylised whole-mission visu-alisation (launch → free-return → splashdown)	A frame-by-frame log of the on-screen numeric dials during re-entry
Re-entry segment length	~3 seconds of screen time (~00:26 → 00:29 of a 30 s clip)	193 real seconds (video clock 1:28:19 → 1:31:32)
Carries geography?	Yes — Earth, continents, a drawn ground track, a splash-down point	No — not one latitude, longit-ude, or heading value exists in the data
Carries numbers?	No — zero velocity/altitude/ time/coordinate readouts on screen	Yes — velocity, altitude, downrange, mission time, Moon distance per frame
Style	Smooth, continuous, artist-rendered arc	Raw samples with duplicates, a time reversal, and a blackout gap

The single most important fact in this whole comparison: the two sources have **no overlap-ing measurable quantity**. NASA shows where; our dials show how fast / how high / when. Neither can confirm or refute the other’s unique claims. Any statement that “the telemetry matches the NASA animation” (or contradicts it) on trajectory **cannot be proven** — they do not measure the same thing.

2. What is PROVABLE from NASA's animation (Entry → Splashdown)

These are things visibly on screen in NASA's own file. No interpretation:

- **P-N1.** The re-entry portion occupies roughly the final ~3 seconds of the clip (~00:26–00:29).
- **P-N2.** A glowing entry marker approaches Earth **from the west** (out of the North Pacific) and travels generally **west-to-east** across northern latitudes.
- **P-N3.** The depicted **splashdown is in the Pacific Ocean, off the west coast of North America** (California / Baja region).
- **P-N4.** There are **no numeric readouts of any kind** in the animation — no velocity, no altitude, no timestamp, no coordinates, no scale bar.
- **P-N5.** The arc is **smooth and continuous** — it contains no visible duplicated frames, no backward jump, and no blackout gap.

3. What is PROVABLE from our telemetry (the dials)

Directly read or arithmetically derived from `traj_telemetry.json` :

- **P-T1.** 265 frames span video clock **1:28:19 → 1:31:32 = 193 s** of coverage.
 - **P-T2.** Velocity falls **25,064 → 11,650 mph** (peak reading 25,192 mph).
 - **P-T3.** Altitude ("FROM EARTH" dial) goes **28 → 40 → 34 mi** — it **rises 12 mi and then dips**, rather than descending monotonically.
 - **P-T4.** Downrange (integrated from dial velocity) reaches **≈ 948 mi** by the last frame.
 - **P-T5.** Peak deceleration derived from the velocity trace is **≈ 9.16 g** (around frame 203).
 - **P-T6.** **154 unique frames + 111 byte-identical duplicate frames** (109 duplicate-of-previous transitions inside the animation window).
 - **P-T7.** **Two time reversals** — the clock jumps backward at frame 38 (and again at 105) to 1:28:19.
 - **P-T8.** **One blackout gap** — between frame 152 and 153 the clock jumps forward **+110 s** with no data in between.
 - **P-T9.** The data **ends at 34 mi altitude / 11,650 mph** — a craft still travelling at hypersonic speed, **well above the ocean**. Our telemetry **never records a splashdown**.
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4. Differences, ambiguities, and anomalies (the deep dive)

4.1 Splashdown location — a corrected assumption

- **NASA (P-N3):** Pacific, off western North America.
- **Our data:** contains **no location at all** (P-T9) and stops while still aloft.
- **Earlier in this project we had assumed splashdown ~28.5°N 75°W — the Atlantic, off Florida.** That assumption is **contradicted by NASA's own depiction** and was never supported by our dials. **It is retracted.** The honest position is: our data cannot place the splashdown anywhere; NASA's animation places it in the Pacific.
- **Status:** our old Atlantic guess = **wrong/unsupported**; NASA Pacific = **what NASA shows** (we did not independently verify it, but it is their own claim).

4.2 Geography vs. numbers — the unbridgeable gap

- NASA has a ground track but no numbers (P-N4); we have numbers but no ground track (P-T... no geography).
- **Ambiguity:** the direction of travel around the globe, the entry corridor, and the landing coordinates are **UNKNOWN from our telemetry**. Our replica therefore refuses to draw a map and instead plots an **altitude-vs-downrange profile** — the only spatially honest thing our data supports.

4.3 Our data ends before splashdown

- NASA's arc reaches the water (P-N3); ours stops at 34 mi / 11,650 mph (P-T9).
- $\approx$ **the entire final descent (from 34 mi down to sea level) is absent from our telemetry**. Anyone claiming our data "shows the landing" would be wrong.

4.4 The altitude rise — possible skip-entry signature

- Altitude climbs 28 $\rightarrow$ 40 mi then settles to 34 (P-T3). A monotonic fall would be expected for a simple ballistic descent.
- **A rise-then-fall is consistent with a skip / lofted entry** (the vehicle uses lift to climb back out briefly before final descent). **This is an observation, not a confirmed maneuver** — the dials show the altitude numbers; they do not label a "skip." NASA's smooth arc (P-N5) shows no such feature, but it also carries no altitude scale, so it **cannot confirm or deny** it.

4.5 Anomalies present in our data, absent in NASA's

NASA's animation is smooth and continuous (P-N5). Our raw telemetry is not:

- **Time reversals** (P-T7) at frames 38 and 105 — the clock runs backward to the start. Consistent with a **broadcast replay/loop artifact**, not real vehicle motion.
- **Blackout gap** (P-T8) — +110 s missing at frame 153, exactly where altitude reads its 40 mi peak. Whatever happened during those 110 s was **not captured**.
- **111 duplicate frames** (P-T6) — long stretches where the image is byte-identical; the broadcast was static/frozen there.
- **A velocity wobble** near frames 189–211 (readings jitter up and down a few hundred mph, e.g. 13,764 $\rightarrow$ 13,842 $\rightarrow$ 13,721) — consistent with **OCR / dial-read noise**, not real re-acceleration.
- **None of these appear in NASA's animation** because it is an artist-smoothed rendering, not a live data feed.

4.6 Timescale mismatch

- NASA compresses the whole mission into 30 s and the re-entry into ~ 3 s (P-N1); our telemetry covers 193 real seconds of that same re-entry (P-T1). They are **not frame-comparable**; you cannot line up "NASA second 27" with any specific telemetry frame.

5. What our replica (`TrueSiddiquiAnimation.html`) does and does not claim

Does (all from real dials):

- Plots every one of the 265 frames as an **altitude (real) vs. downrange (real)** profile.
- Shows live velocity, altitude, downrange, derived g-force, mission time, and Moon distance.
- Marks the **entry point**, the **40 mi altitude peak** (skip signature), the **time reversal**, the **blackout gap**, and the **hard data-end** at 34 mi / 11,650 mph.

Does NOT (honest omissions):

- Does **not** draw a map, a ground track, continents, or a splashdown point — we have no geography.
- Does **not** continue the trajectory to the ocean — our data stops while the craft is still aloft.
- Does **not** smooth away the duplicates, reversal, or gap — they are shown as they are.

6. Bottom line

- **NASA's animation and our telemetry measure different things and cannot verify each other.** NASA = geography, no numbers. Us = numbers, no geography.
- **NASA depicts a Pacific splashdown.** Our data neither confirms nor denies it — and our earlier Atlantic/Florida assumption is **retracted as unsupported**.
- **Our telemetry stops at 34 mi / 11,650 mph, still aloft** — it contains **no splashdown**, so the final descent is simply missing from our record.
- **Our data carries real anomalies** (2 time reversals, a 110 s blackout, 111 duplicate frames, read-noise wobble) that NASA's smooth animation does not — because one is a raw capture and the other is a stylised rendering.
- **The altitude 28→40→34 profile is a genuine, provable feature** of our data and is consistent with a skip/lofted entry, but the dials do not label it as such, so it stays an **observation, not a claim**.

Truth, nothing but truth: where the data could not prove something, this document says so instead of guessing.